SHIELD-W: When does drought-tolerant vegetation protect urban cooling under a finite water budget?

Soil–Hydraulic Integration for Evaluating Leaf cooling under Drought: Water-budget extension

CM617 Sustainability Research Project

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Abstract

Selecting vegetation using cooling at a prescribed soil water potential can obscure how rapidly plants consume their available water. We extend the supplied SHIELD prototype with a mass-balanced root-zone bucket, conservative shortwave interception, a robust discrete physiological search, and an exact update for the linear atmospheric transport operator. Controlled simulations compare hydraulic strategy, leaf area index (LAI), irrigation budget and irrigation timing over a seven-day synthetic hot spell. At LAI 3 and prescribed dry soil, the safe–conservative (SC) archetype cools the 2 m air by 3.88 K, compared with 2.60 K for the efficient–vulnerable (EV) archetype. Under a 12 mm seven-day daily-irrigation budget, however, their final-day cooling is 1.84 and 1.91 K, respectively. The distinction is explained by water depletion rather than a contradiction in hydraulic vulnerability. Total cooling resilience also exceeds transpiration-increment resilience because shade persists. These are original comparative computations for this course project, not field-validated forecasts or a claim of a globally new ecohydrological model. All figures, tables, assumptions and executable code accompany the report.

1 Question, prior work and specific contribution

Urban planting must sustain cooling when heat and water scarcity coincide. Existing research already couples urban radiation, vegetation physiology and hydrology: UT&C is an established example [1]. Soil moisture, urban form and the division between shading and evaporation have also been investigated explicitly [2]. Consequently, simply adding soil moisture or comparing shade with transpiration is not a defensible literature-wide novelty claim.

The supplied SHIELD project used a P-Hydro-inspired leaf calculation within a one-dimensional air column, but prescribed soil potential independently for each day. Its missing comparison is operational: **does the trait ranking at identical soil water potential persist when both plants start with identical stored water and receive the same irrigation allowance?** We answer this question using a reproducible finite-water experiment, and assess whether the answer depends on numerical resolution or selected model assumptions.

P-Hydro motivates the hydraulic-profit structure [3]; the present Python closure is a reduced adaptation of the supplied prototype. It is neither the official P-Hydro solver nor a reproduction of its calibration. The documented P-Hydro and UT&C repositories provide transparent reference implementations [5, 6]; they were consulted, not installed or executed. Environmental-biophysics theory supplies the physical basis for leaf heat and vapour exchange [4]. The soil retention law follows van Genuchten [7], with loam class-average coefficients from Carsel and Parrish [8].

The course-level original contributions are the paired fixed-potential/finite-storage comparison, irrigation-budget and timing experiments, independent hydraulic-trait ablation, an explicit test of shade masking in resilience ratios, and numerical verification. None of the individual component equations is claimed as new.

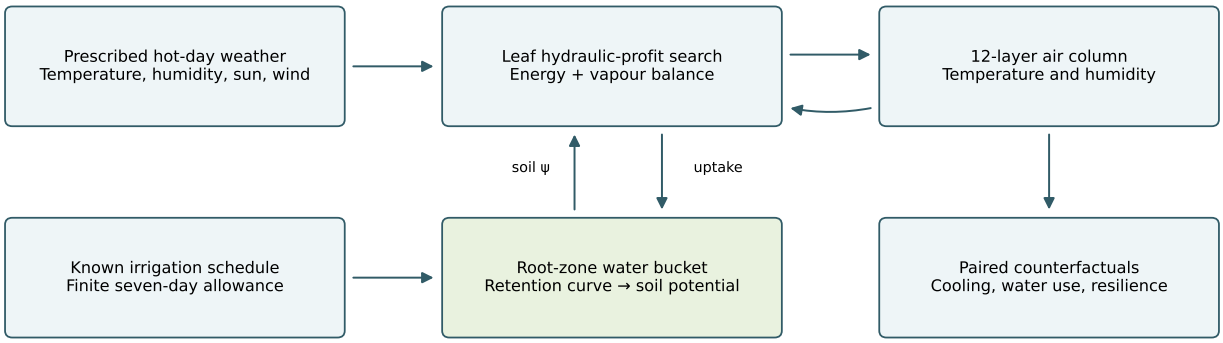


Figure 1: Implemented feedbacks. Background weather is prescribed. Leaf transpiration depletes the root-zone bucket, which changes the potential supplied to the next leaf calculation. Air temperature and humidity feed back onto leaf vapour demand. Irrigation enters as a known daily impulse. All plots are generated from this project's simulations.

2 Mathematical model and implementation

2.1 Domain, units and forcing

The horizontally homogeneous air column is 12 m deep, discretized into 12 finite-volume cells. Temperature at exactly 2 m is obtained by linear interpolation between cell centres. LAI is one-sided leaf area per ground area. For $z_i = (i + 1/2)\Delta z$, the leaf-area density is proportional to $(z_i/H)^{1.4}(1 - z_i/H)$ and normalized so that $\sum_i a_i \Delta z = \text{LAI}$. This idealized profile is an analyst choice, not a measured street-tree crown.

Each day repeats the same controlled forcing, with h the local hour:

$$p(h) = \sin[2\pi(h - 9)/24], \tag{1}$$

$$T_{bg}(h) = 273.15 + 31 + 5.5p(h) + \delta T, \tag{2}$$

$$RH_{bg}(h) = \text{clip}[0.58 - 0.18p(h), 0.28, 0.90], \tag{3}$$

$$S_0(h) = 900 \max\{0, \sin[\pi(h - 6)/12]\} \quad (6 \le h \le 18), \tag{4}$$

$$u_{top}(h) = 1.8[0.75 + 0.35 \max\{p(h), 0\}]. \tag{5}$$

Outside daylight $S_0 = 0$. Here T is kelvin, S_0 is W m^{-2} and u is m s^{-1}. Default $\delta T = 0$; sensitivity tests use 3 K warming at the same relative-humidity schedule, so absolute atmospheric moisture also changes. This is a synthetic forcing ensemble, not downloaded meteorology. Pressure is 101325 Pa, $c_p = 1005$ J kg^{-1} K^{-1} and $M_w = 0.01801528$ kg mol^{-1}. A one-day atmospheric spin-up at the initial prescribed soil potential precedes the seven scored days; spin-up does not consume the scored initial water store.

2.2 Hydraulic supply and reduced profit closure

Remaining conductivity is represented by

$$v(\psi) = 2^{-|\psi/\psi_{50}|^b}, \qquad E_h = K_p \int_{\psi_s - \Delta\psi}^{\psi_s} v(\psi)\, d\psi. \tag{6}$$

K_p has units mol H_2O m$^{-2}_{leaf}$ s^{-1} MPa^{-1}, E_h is mol m$^{-2}_{leaf}$ s^{-1} and potentials are MPa. A 16-point Gauss–Legendre quadrature evaluates this integral. Effective path length and viscosity are absorbed into K_p; root conductivity, soil-to-root resistance, hydraulic capacitance and persistent embolism are absent.

Candidate transpiration is imposed on the energy balance, yielding T_L and the leaf-to-air normalized vapour deficit

$$D_L = \frac{e_s(T_L) - e_a(q)}{P}, \quad g_{tw} = E_h/D_L, \quad g_{sw} = \left(g_{tw}^{-1} - g_{bw}^{-1}\right)^{-1}. \tag{7}$$

Thus hydraulic supply equals atmospheric water demand by construction for feasible candidates. The specific-humidity conversion is $e_a = qP/(0.622 + 0.378q)$. The Buck-type saturation approximation implemented in the code is

$$e_s(T) = 611.21 \exp\left[\left(18.678 - \frac{T_C}{234.5}\right)\frac{T_C}{257.14 + T_C}\right], \quad T_C = T - 273.15. \tag{8}$$

Reject candidates with $D_L \le 10^{-6}$, $g_{tw} \ge 0.98 g_{bw}$, or $g_{sw} > 1.2$ mol m^{-2} s^{-1}. These bounds are computational/physiological assumptions. The CO_2 conductance is

$$g_{tc}^{-1} = (g_{sw}/1.6)^{-1} + (g_{bw}/1.37)^{-1}, \quad A = g_{tc} c_a (1-\chi), \tag{9}$$

where $c_a = 420$ ppm makes A numerically μmol CO_2 m^{-2} s^{-1}. For each candidate internal ratio $\chi = c_i/c_a$, define

$$f_J = \frac{c_i(1-b_r) - (\Gamma^* + b_r K_m)}{c_i + 2\Gamma^*}, \quad J = 4A/f_J, \tag{10}$$

$$J_{light} = 4\phi_0 I, \quad J_{max} = \frac{J_{light}}{\sqrt{(J_{light}/J)^2 - 1}}, \tag{11}$$

$$F = A - \alpha_c J_{max} - \gamma_c (\Delta\psi)^2. \tag{12}$$

Here $b_r = 0.002$, $\phi_0 = 0.087$, $I = 2.05 S_{intercepted}$ is the prescribed effective photon conversion, $\alpha_c = 0.08$ is dimensionless and $\gamma_c = 1$ has the units required to express hydraulic cost as a carbon rate per MPa^2. No independent carboxylation-capacity ceiling is imposed. Reject $f_J \le 0$, $J \le 0$ or $J \ge 0.995 J_{light}$. The zero-transpiration, zero-profit state is explicitly available.

For temperature response,

$$k(T) = k_{25} \exp\left[\frac{E_a(T - 298.15)}{298.15 RT}\right].$$

The (k_{25}, E_a) pairs are $(42.75, 37830)$ for Γ^*, $(404.9, 79430)$ for K_c and $(278400, 36380)$ for K_o, with $K_m = K_c(1 + 210000/K_o)$. These constants and the carbon-cost closure are inherited prototype assumptions, not newly calibrated coefficients. $R = 8.314462618$ J mol^{-1} K^{-1}.

The search uses 81 quadratically spaced $\Delta\psi$ candidates in $(0, 6]$ MPa and 61 equally spaced χ values in $[0.35, 0.95]$. It returns the best feasible discrete profit, not a guaranteed continuous global optimum. Unlike the supplied code, the common drawdown domain does not impose a trait-dependent leaf-potential cutoff. Resolution tests use both coarser and finer search grids. Capacity is recalculated at every step: **this is an equilibrium acclimation approximation**. Actual P-Hydro distinguishes short-term stomatal responses from slower capacity acclimation [3]; the present model does not reproduce that timescale separation. No nighttime or cuticular transpiration is included.

2.3 Leaf energy, wind and conservative shortwave interception

The steady, two-sided energy balance is

$$\begin{aligned} 0 = {} & \alpha_{SW} S_{intercepted} + \epsilon\sigma(T_{sky}^4 + T_g^4 - 2T_L^4) \\ & - 2h_{one}(T_L - T_a) - \lambda(T_L) M_w E, \end{aligned} \tag{13}$$

$$\lambda(T_L) = 2.501 \times 10^6 - 2361(T_L - 273.15). \tag{14}$$

Use $\alpha_{SW} = 0.72$, $\epsilon = 0.96$, $T_{sky} = T_{bg} - 12$ K and $T_g = T_{bg} + 8(S_0/900)\exp(-k\mathrm{LAI})$. The Stefan–Boltzmann constant is $5.670374419 \times 10^{-8}$ W m^{-2} K^{-4}. Twelve vectorized Newton iterations solve the energy balance; the maximum residual is saved. Leaf heat storage is neglected. The imposed sky/ground temperatures do not form a closed whole-column radiative budget.

For leaf dimension $L_c = 0.06$ m and wind $u_i = \max[0.05, u_{top}\exp(-0.28(L_i^{\uparrow} + \ell_i/2))]$,

$$Re = \frac{u_i L_c}{\nu}, \quad Nu = 0.664 Re^{1/2} Pr^{1/3}, \quad Sh = 0.664 Re^{1/2} Sc^{1/3}, \tag{15}$$

$h_{one} = Nu\, k_a/L_c$ and $g_{bw} = (ShD_v/L_c)P/(RT_a)$. The code gives temperature-dependent air viscosity, conductivity and vapour diffusivity explicitly. This is a forced-convection approximation; leaf flutter, free convection and real crown aerodynamics are not resolved.

Set $\ell_i = a_i \Delta z$ and $L_i^{\uparrow} = \sum_{j>i} \ell_j$, with $k = 0.55$. Intercepted shortwave per leaf area is

$$S_{intercepted,i} = \frac{S_0 e^{-kL_i^{\uparrow}}(1 - e^{-k\ell_i})}{\ell_i}. \tag{16}$$

This conserves intercepted plus transmitted radiation across discrete layers. The remainder associated with absorptivity is not tracked through multiple scattering. Ground sensible heating is prescribed as $H_g = 300(S_0/900)e^{-k\mathrm{LAI}}$ W m^{-2}. These prescriptions isolate canopy comparisons; they do not predict a pavement energy balance. In the bare control LAI is zero. In the shade control LAI is unchanged and transpiration is disabled, but leaves still exchange sensible heat and longwave radiation. Therefore "shade" is shorthand for a *non-transpiring canopy counterfactual*, not a pure radiative shade intervention.

2.4 Atmospheric transport and feedback

For specific humidity q (kg kg^{-1}), air density $\rho = P/[287.05T_a(1 + 0.61q)]$, and turbulent diffusivity $K(z) = 0.35 + (z/H)^{1.3}$ m^2 s^{-1},

$$\partial_t T_a = \partial_z(K\partial_z T_a) - r(z)(T_a - T_{bg}) + \frac{aH_L}{\rho c_p} + \frac{H_g \mathbf{1}_{i=0}}{\rho c_p \Delta z}, \tag{17}$$

$$\partial_t q = \partial_z(K\partial_z q) - r(z)(q - q_{bg}) + \frac{aM_w E}{\rho}, \tag{18}$$

$$r_i = [150(1.25 - 0.5z_i/H)]^{-1} + \mathbf{1}_{i=N-1}/180. \tag{19}$$

Zero diffusive boundary fluxes apply at the top and bottom; relaxation supplies lateral/background exchange and additional top entrainment. There is no explicit horizontal urban canyon, no direct latent-heat sink in the air equation and no double counting of evaporation: latent cooling enters through the reduced leaf sensible heat H_L. Humidity is predicted rather than prescribed inside the canopy, and is checked for supersaturation without silent clipping. The ground is assumed mulched/non-evaporating so the counted water loss is entirely plant transpiration; rainfall, surface interception and direct soil evaporation are zero. This is a deliberate experimental simplification, not a universal description of urban green space.

2.5 New finite-water bucket and irrigation

Root-zone storage per ground area is $W = 1000Z_r\theta$ mm, where $Z_r = 0.6$ m and θ is volumetric water content. The retention relationship is

$$\theta(\psi) = \theta_r + (\theta_s - \theta_r)[1 + (\alpha|h|)^n]^{-m}, \quad m = 1 - 1/n, \quad |h| = |\psi|/0.00980665. \tag{20}$$

The inverse supplies ψ_s at each step. Loam values are $\theta_r = 0.078$, $\theta_s = 0.43$, $\alpha = 3.6$ m^{-1} and $n = 1.56$ [8]. The α conversion from 0.036 cm^{-1} matters. Initial $\psi_s = -0.15$ MPa gives $\theta_0 = 0.1153$. Initial storage above the default bookkeeping floor $\psi_{floor} = -6$ MPa is 19.53 mm. This floor is *not* a measured wilting point or a mortality threshold.

At each step, irrigation is added first; excess above θ_s drains immediately. Subsequent canopy water use is

$$U_n = M_w \Delta t \sum_i E_i \ell_i, \qquad 1000Z_r(\theta_{n+1} - \theta_n) = I_n - U_n - Q_n. \tag{21}$$

Numerically one kg m^{-2} of water equals one mm. There is no unsaturated drainage below saturation, capillary recharge, root redistribution or infiltration delay. A candidate cap $E_i \le W_{available}/(M_w \mathrm{LAI}\Delta t)$ prevents total uptake from exceeding the available store. This conservative uniform per-leaf cap is not an optimal allocation among canopy layers. It is relevant near depletion and is audited by varying the storage floor.

All policies are evaluated at 06:00 daily. With budget B, front-loading applies B on day 1; the daily policy applies $B/7$ each day; the threshold policy applies up to 4 mm whenever the pre-irrigation $\psi_s < -0.8$ MPa, subject to remaining budget. A zero-budget control receives no irrigation. Threshold management is a fixed heuristic, not a numerically optimized controller. Delivered water, rather than allowance alone, is reported; default timing cases deliver 12 mm to the threshold runs. Irrigation efficiency is assumed 100%, so results concern water entering the root-zone bucket.

2.6 Numerical correction and diagnostics

The previous source-then-relax update for a scalar anomaly x was $x_{n+1} = e^{-\Delta t/\tau}(x_n + \Delta t s)$. Its stationary response is $s\Delta t/(e^{\Delta t/\tau} - 1)$ rather than the correct $s\tau$. At $\Delta t = 180$ s and $\tau = 150$ s, their ratio is 0.517. This is a mathematical illustration of source suppression, not a direct correction factor for the nonlinear full model.

For the revised discretized operator $\mathcal{A} = \mathcal{L} - \mathrm{diag}(r)$, compute once

$$M = e^{\mathcal{A}\Delta t}, \quad B = \mathcal{A}^{-1}(M - I), \quad x_{n+1} = x_{bg} + M(x_n - x_{bg}) + B\, s_n. \tag{22}$$

This update is exact only for the linear column problem with background and sources held constant during the step; the nonlinear leaf coupling still requires timestep checks. Sources are refreshed every step, with midpoint weather and post-step air outputs. Default $\Delta t = 600$ s. Every record represents exactly Δt seconds, and integrated water use is summed from actual step amounts. By contrast, the supplied code recorded every $3 \times 180 = 540$ s but multiplied output water rates by 600 s, inflating that particular integration by $600/540 - 1 = 11.1\%$ for its regular output grid.

3 Experimental design and performance metrics

Experiment	Factors	Purpose
Static factorial	2 traits × 3 LAI × 2 soil potentials	Revisit earlier comparison
Finite-water factorial	2 traits × 3 LAI × 4 budgets	Joint water/cooling trade-off
Timing comparison	2 traits × 3 schedules; LAI 3, $B = 12$	Allocation through time
Independent traits	2 K_p × 2 ψ_{50}; LAI 3, $B = 12$	Separate correlated traits
Sensitivity	36 environmental/storage settings, paired traits	Conditional robustness
Cutoff audit	3 soil cutoffs × 2 traits	Identify imposed-floor effects
Numerics	Timestep, height resolution, search resolution	Estimate numerical sensitivity

Table 1: Executed experiments. Bare and shade counterfactuals accompany comparisons. The timing comparison adds two schedules beyond the daily-budget factorial.

The EV archetype uses $(K_p, \psi_{50}, b) = (0.008, -1.4, 1.5)$; SC uses $(0.0055, -3.0, 1.5)$. The label "conservative" refers to lower maximum conductance, not a guarantee of lower water use under dry conditions. Neither archetype is an identified species. Static soils are -0.3 and -1.8 MPa. LAI values are 1, 3 and 5; budgets are 0, 6, 12 and 24 mm over seven days.

Define signed cooling $\Delta T = T_{bare,2m} - T_{full,2m}$ and the conditional transpiration increment $\Delta T_E = T_{shade,2m} - T_{full,2m}$. Positive values mean cooler air. They satisfy $\Delta T = (T_{bare} - T_{shade}) + \Delta T_E$, an exact counterfactual arithmetic identity, not a causal attribution independent of canopy feedbacks. Report the average over intervals spanning 14:00–16:00 and seven-day daytime service

$$C = \sum_{d=1}^{7} \int_{10}^{18} \Delta T_d(h)\, dh, \quad C_E = \sum_{d=1}^{7} \int_{10}^{18} \Delta T_{E,d}(h)\, dh. \tag{23}$$

Both are in K h; no negative values are truncated. They are local air-temperature integrals, not citywide cooling energy or human thermal-comfort indices. Marginal irrigation benefit is $\eta = [C(B) - C(0)]/I_{delivered}$, defined only for positive delivered water.

Two static resilience ratios are

$$CRI_T = \Delta T_{dry}/\Delta T_{wet}, \qquad CRI_E = \Delta T_{E,dry}/\Delta T_{E,wet}. \tag{24}$$

Ratios are only interpretable when denominators are positive and appreciable. If shade contribution is $S > 0$ and wet/dry increments are $E_w > E_d \geq 0$, then $(S + E_d)/(S + E_w) > E_d/E_w$: the shade floor mathematically raises the apparent total resilience. These ratios are diagnostic definitions for this analysis, not a claim that resilience ratios are a new invention.

Sensitivity crosses $Z_r \in \{0.3, 0.6, 1.2\}$ m, $\tau \in \{75, 150, 300\}$ s, initial $\psi_s \in \{-0.10, -0.30\}$ MPa and $\delta T \in \{0, 3\}$ K. The 36 settings are equally enumerated scenarios, **not random draws, confidence intervals or occurrence probabilities**. Traits remain paired under the same setting. Numerical tests vary timestep from 600 to 150 s, cells from 12 to 24, and physiological search resolution. An additional experiment reintroduces the old source splitting while holding revised physics fixed to isolate the integration effect. All runs are deterministic; no random seeds or statistical significance tests apply.

4 Results with physical interpretation

4.1 Static comparison and the importance of shade masking

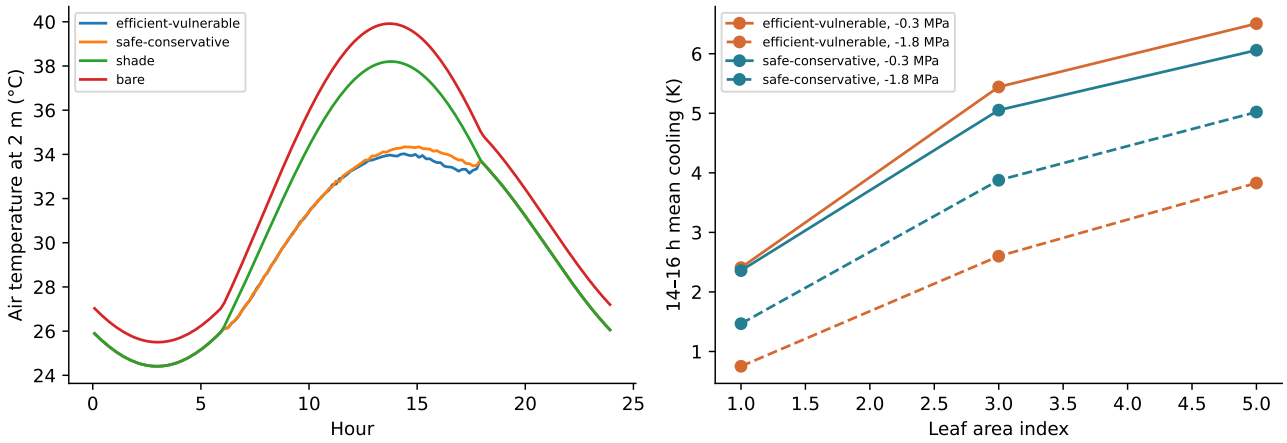


Figure 2: Left: one-day wet-soil diurnal air temperatures for bare, non-transpiring canopy and the two full models at LAI 3. Right: 14–16 h cooling in the static factorial. Solid lines denote −0.3 MPa and dashed lines −1.8 MPa. Colours distinguish archetypes.

At LAI 3, wet-soil cooling is 5.44 K for EV and 5.05 K for SC. Dry-soil cooling is 2.60 and 3.88 K. Greater EV conductivity permits a stronger well-watered latent flux. At the same dry potential, however, SC retains more conductivity and can sustain more transpiration. These are controlled state comparisons: the dry-soil experiment has an implicit external reservoir maintaining its prescribed potential despite water extraction.

At LAI 3, total CRI is 0.48 for EV and 0.77 for SC, while transpiration-increment CRI is only 0.25 and 0.66. In particular, a large total ratio does not mean that the same proportion of evaporative cooling survives. Increasing LAI intercepts more radiation and reduces prescribed ground heating, so shade raises the ratio even when plant water loss declines. The increment is conditional on a non-transpiring canopy and includes the resulting changes to leaf and air states.

The old numerical values (including its 3.05 K reference result) are retained only in `legacy/` as supplied historical outputs. They should not be mixed with the revised figures. The revised setup changes transport integration, discrete radiation interception, ground-shading consistency, the physiological search, potential-domain constraints, spin-up, exact 2 m interpolation and averaging windows. Thus differences cannot be assigned entirely to a single correction.

4.2 Finite water changes the late-heatwave comparison

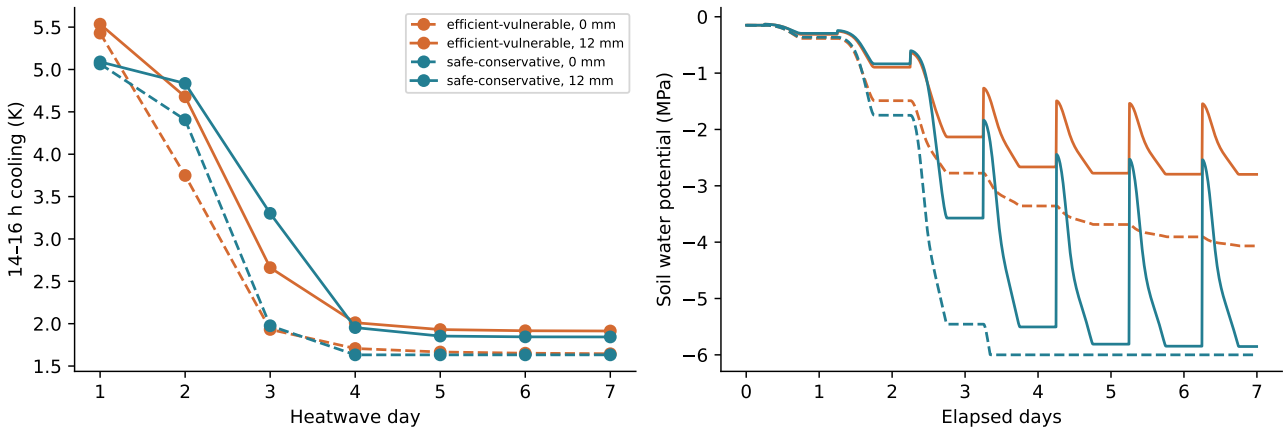


Figure 3: Seven-day dry-down at LAI 3. Dashed lines receive no irrigation; solid lines receive 12 mm in daily portions. Left: afternoon cooling. Right: the evolving soil potential; irrigation causes upward jumps. Both archetypes start with the same root-zone store.

With daily 12 mm irrigation, C is 165.67 K h for EV and 168.36 K h for SC, but final-day cooling is 1.91 and 1.84 K. Total transpiration is 30.01 and 31.49 mm. The finite bucket therefore reveals a distinction between accumulated service and the remaining afternoon service: a plant can cool more earlier, use more stored water, and lose that advantage later.

This is a feedback explanation, not an additional fitted mechanism. Maintaining conductivity at low potential permits continued extraction, which drives the same finite soil volume to a more negative potential. The fixed-soil comparison suppresses precisely that feedback. Differences near the end of the simulation must also be assessed against the soil-cutoff audit and numerical changes below. The model does not calculate tree survival, so lower cooling cannot be equated with greater mortality or poorer species suitability.

Trait	Schedule	I (mm)	E (mm)	C (K h)	Day 7 (K)
EV	none	0.00	18.84	140.57	1.65
EV	daily	12.00	30.01	165.67	1.91
EV	front	12.00	30.73	175.56	1.65
EV	threshold	12.00	30.56	167.44	1.68
SC	none	0.00	19.53	142.14	1.63
SC	daily	12.00	31.49	168.36	1.84
SC	front	12.00	31.53	177.07	1.63
SC	threshold	12.00	31.53	170.97	1.63

Table 2: LAI 3 timing and water comparison. EV and SC denote the two archetypes. I is delivered irrigation, E is cumulative transpiration, C is seven-day 10–18 h cooling service and Day 7 is the final 14–16 h mean. Transpiration can exceed irrigation because plants also consume initially stored water.

4.3 Budget, LAI and timing create different objectives

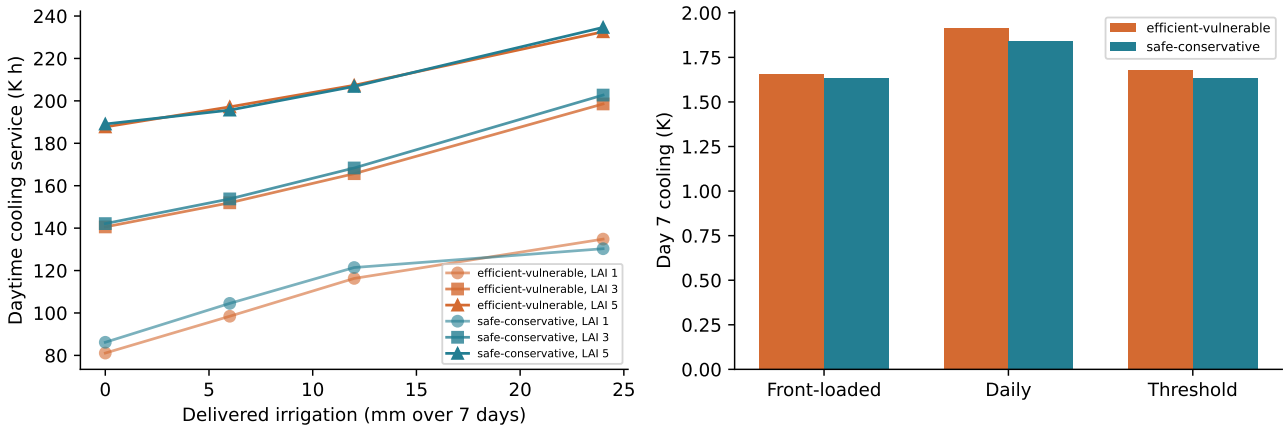


Figure 4: Left: cooling service against delivered irrigation for daily policies and three LAIs. Curves are sampled response curves, not an optimized Pareto frontier. Right: final-day cooling under equal 12 mm allowances and three schedules at LAI 3. The CSV reports actual delivery for every policy.

At LAI 3, front-loading yields 175.56 and 177.07 K h, whereas daily delivery yields 165.67 and 168.36 K h. Front-loading makes water available during the initial high-conductance period; this can increase early transpiration and accumulated cooling. Daily delivery reserves part of the allowance for later days and yields final-day cooling of 1.91 and 1.84 K, compared with 1.65 and 1.63 K for front-loading. Consequently, a schedule preferred for cumulative cooling need not be preferred for heatwave-end protection. This is a comparison of three specified schedules, not proof that any schedule is globally optimal.

For daily 12 mm irrigation, incremental benefit relative to the corresponding zero-irrigation canopy is 2.09 K h mm^{-1} for EV and 2.19 for SC. This excludes the zero-irrigation shade benefit from the numerator. The denominator is water actually delivered to the root zone, with no pumping energy, irrigation losses or monetary valuation. Increasing LAI gives more interception but also more potential leaf-area water demand; the curves expose both effects without implying that the highest tested LAI is a universal planning optimum.

4.4 Sensitivity and independent trait checks

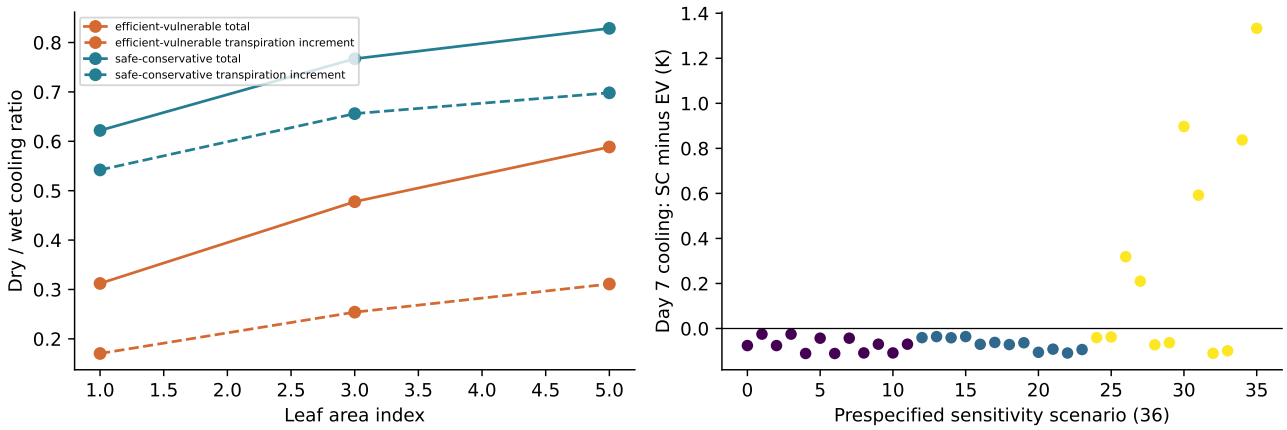


Figure 5: Left: total and transpiration-increment static resilience. Right: the paired SC minus EV final-day cooling difference across 36 prespecified settings; positive values favour SC for this metric. Scenarios are ordered by root depth, relaxation time, initial potential and warming. The colour changes with root depth and does not encode a probability.

Across the 36 sensitivity settings, SC has greater final-day cooling in 6 cases, EV in 30 cases, with differences ranging from -0.11 to 1.33 K. These counts describe only the selected design. The paired result depends on both atmospheric exchange and the time history of storage. Large root zones dilute the fractional depletion caused by a given extraction, whereas stronger ventilation weakens local temperature anomalies. A change in initial water content modifies how much transpiration can occur before hydraulic limitation becomes strong.

The independent 2×2 trait ablation prevents the original correlated archetypes from being interpreted as an isolated test of drought tolerance. At fixed $K_p = 0.0055$, changing ψ_{50} from -1.4 to -3.0 MPa changes C from 163.35 to 168.36 K h and final-day cooling from 1.98 to 1.84 K. At fixed $K_p = 0.008$, changing ψ_{50} from -1.4 to -3.0 MPa changes C from 165.67 to 166.50 K h and final-day cooling from 1.91 to 1.63 K. The response is therefore an interaction between water transport and the evolving soil store, rather than an effect that can be assigned solely to the archetype label. Full values are in `independent_traits.csv`.

With the SC bookkeeping floor moved from -6 to -4 MPa, final-day cooling changes from 1.84 to 1.63 K; moving it to -8 MPa gives 1.84 K. The default SC run ends at -5.85 MPa. Thus the LAI 3 daily-irrigation comparison is insensitive to extending the floor to -8 MPa, but a shallower allowable store changes the answer. This sensitivity is reported rather than treating the chosen floor as a biological fact. See `cutoff_audit.csv`.

4.5 Numerical verification and its limits

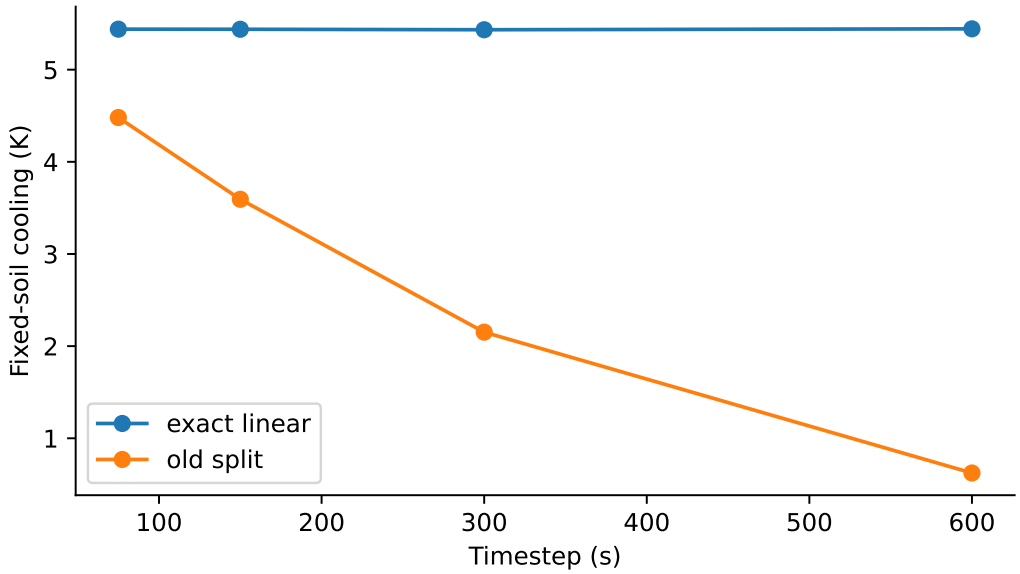


Figure 6: Controlled integration audit with revised physics and static $\psi_s = -0.3$ MPa. The old source-then-relax scheme increasingly suppresses cooling at longer timesteps. The exact linear transport update removes that particular bias, while leaf coupling still has discretization error. These are new audit simulations, not reruns of the complete old configuration.

For efficient-vulnerable, reducing the timestep from 600 to 300 to 150 s gives $C = 165.67, 165.86, 166.01$ K h and final-day cooling $1.914, 1.906, 1.902$ K. At 150 s, doubling the vertical grid to 24 cells gives $C = 168.59$ K h and final-day cooling 1.924 K. At 300 s and 12 cells, the finer physiological search gives $C = 165.57$ K h and final-day cooling 1.904 K.

For safe-conservative, reducing the timestep from 600 to 300 to 150 s gives $C = 168.36, 168.62, 168.80$ K h and final-day cooling $1.843, 1.836, 1.832$ K. At 150 s, doubling the vertical grid to 24 cells gives $C = 171.46$ K h and final-day cooling 1.853 K. At 300 s and 12 cells, the finer physiological search gives $C = 168.31$ K h and final-day cooling 1.836 K.

These comparisons quantify discretization sensitivity for the selected dynamic cases, rather than establishing convergence for every scenario. Small trait differences and close schedule rankings should be interpreted against these changes. See `convergence.csv` for all tested resolutions.

Across 249 executed model calls, maximum absolute bucket residual is 2.3e-13 mm, maximum leaf-energy residual is 1.8e-12 W m^{-2} and maximum model relative humidity is 0.973. These are *numerical consistency checks*, not observations and not model validation. Eleven executable checks cover the soil-retention inverse, discrete shortwave accounting, forced-column equilibrium, trait vulnerability ordering, coupled water/leaf-energy balance, and the zero-transpiration control, together with independent scalar energy roots and hydraulic-profit search, an independent atmospheric ODE solution, per-step storage accounting and interval-independent metrics. No field accuracy score can be calculated without independent observations.

5 Assumptions, limitations and what may be concluded

Assumption or omission	Consequence for interpretation
Synthetic repeating weather and no rainfall	Isolates mechanisms; does not represent an observed heatwave or climate distribution.
Archetype hydraulic traits and inherited carbon costs	No species rankings; parameter plausibility is not calibration. The sensitivity grid does not span uncertainty in every physiological coefficient.
Equilibrium capacity at each timestep	Ignores acclimation lag and may change subdaily water use. Absolute rates are conditional on this closure.
Reversible vulnerability and no plant damage	Rewetting restores the model instantly; survival, embolism recovery and leaf shedding are not predicted.

Assumption or omission	Consequence for interpretation
Single well-mixed root zone	Ignores vertical water gradients, soil–root resistance and groundwater. A no-unsaturated-drainage bucket can retain early irrigation too effectively.
Fixed canopy during drought	Shade persists even at severe water stress. Leaf loss could reduce the inferred shade floor.
Prescribed longwave environment and ground heat	The leaf energy balance closes numerically, but the whole urban surface/atmosphere energy budget is not closed.
One-dimensional air and prescribed exchange	No buildings, radiation trapping, street orientation, resolved turbulence or off-site heat transport. Absolute cooling is sensitive to ventilation assumptions.
No direct soil evaporation	Isolates plant water use; cannot compare irrigated turf, bare wet soil or permeable pavement fairly.
Air temperature as sole service outcome	Omits mean radiant temperature, humidity-dependent heat stress, wind exposure and building loads. No UTCI or health-benefit claim follows.

The project supports a conditional conclusion: **equal water potential and equal water supply are different experimental questions**. A hydraulic trait that preserves transpiration at equal drought stress can accelerate depletion at equal initial storage. A total cooling ratio also includes a structural shade contribution. These insights justify evaluating time-resolved water and cooling jointly before translating a model ranking into planting advice.

A useful validation sequence would first fit leaf fluxes and hydraulic traits with independent physiological data, then evaluate measured transpiration and canopy temperature, and finally test pedestrian-level air temperature and humidity at a site with measured irrigation and root-zone water content. Parameter fitting and validation should use separate periods. Measurements should constrain ventilation and surface radiation as well as plants; otherwise temperature agreement can mask compensating errors. A comparison with the official P-Hydro solver and a validated urban model such as UT&C would then test how strongly the conclusions depend on the reduced closure.

6 Reproducibility, provenance and course deliverables

From the extracted project directory, run

```
python -m pip install -r requirements.txt
python -m unittest test_model.py -v
python run_experiments.py
python build_report_assets.py
latexmk -pdf report_detailed.tex
latexmk -pdf report_short.tex
```

The completed run's exact package versions and source SHA-256 hashes are in `results/metadata.json`. Runtime is machine dependent. All simulation CSVs, a run-level audit, PNG and vector-PDF figures, BibTeX references, numerical checks and report assets are included. Default simulations are deterministic and need no external files or internet after installing dependencies. `legacy/` contains supplied CSVs only; the revised solver does not ingest them. Source code derives from the user-supplied SHIELD prototype and is substantially rewritten for the present extension. The bibliography identifies scientific sources; neither external repository is bundled.

The two-page main report and this detailed supplement use pdfLaTeX; reference entries are included directly in the LaTeX source. The supplement provides the full methods, numerical audits and additional results accompanying the main report.

AI assistance was used to inspect the prototype, develop the extension, run numerical experiments and prepare the report.

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Team contributions

All four team members contributed equally to this project.